

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Introduction

The Requirement Analysis phase defines the functional and technical requirements of the project. The main purpose is to identify how users, roles, tables, and Access Control Lists (ACLs) will work together to control record access.

The project uses scripted ACLs to determine whether a user can perform specific operations on a record based on the value of a particular field.

2. User and Role Requirements

The system requires users and roles to be created before implementing the ACLs.

Users will be assigned appropriate roles, and these roles will be used as part of the access-control mechanism. This provides a basic structure for identifying authorized and unauthorized users.

Requirements:

- Create the required users.
- Create the required roles.
- Assign appropriate roles to users.
- Use roles as part of the ACL conditions.

3. Table Requirements

A table is required to store the records that will be protected by the ACLs.

The table should contain the required fields, including the field whose value will be evaluated by the ACL script.

Requirements:

- Create the required table.
- Define the necessary fields.
- Add the field used for access validation.
- Create sample records for testing.

4. ACL Requirements

The project requires four different ACL operations. Each ACL will control a specific type of record access.

4.1 Read ACL

The Read ACL will control whether a user can view a record.

The ACL script will check the required field value and determine whether the access condition is satisfied.

Expected behavior:

Condition Satisfied → Read Access Allowed

Condition Not Satisfied → Read Access Restricted

4.2 Create ACL

The Create ACL will control whether a user can create a new record.

The user's role and the required access conditions will be evaluated before allowing the record to be created.

Expected behavior:

Authorized User → Record Creation Allowed

Unauthorized User → Record Creation Restricted

4.3 Write ACL

The Write ACL will control whether a user can modify an existing record.

The ACL will evaluate the defined conditions before allowing changes to the record.

Expected behavior:

Condition Satisfied → Modification Allowed

Condition Not Satisfied → Modification Restricted

4.4 Delete ACL

The Delete ACL will control whether a user can delete an existing record.

The ACL will validate the required access conditions before allowing the deletion operation.

Expected behavior:

Authorized User → Deletion Allowed

Unauthorized User → Deletion Restricted

5. Milestone Requirements

The project implementation is divided into seven milestones:

Milestone Requirement	Purpose
1	Creation of Users and Roles
2	Creation of Tables
3	Creation of ACL – Read
4	Creation of ACL – Create
5	Creation of ACL – Write
6	Creation of ACL – Delete
7	Conclusion
	Establish user-based access control
	Store and manage project records
	Control record viewing
	Control record creation
	Control record modification
	Control record deletion
	Test, verify, and document the implementation

6. Overall Access-Control Flow

The expected access-control flow is:

User Login

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User Role Identified

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User Attempts an Operation

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Relevant ACL is Triggered

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Field Value and Conditions are Checked

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Access Decision is Made

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Operation Allowed or Restricted

7. Expected Outcome

After satisfying the requirements, the project should be able to control different record operations using scripted ACLs.

The system should demonstrate that:

- Users and roles can be configured for access control.
- Records can be stored in the required tables.
- Read access can be controlled.
- Record creation can be controlled.

- Record modification can be controlled.
- Record deletion can be controlled.
- Field values can be used as part of the access-control logic.

8. Phase 2 Conclusion

The Requirement Analysis phase establishes the requirements for implementing the Script-Controlled ACL project. The seven milestones provide a structured approach for creating users and roles, designing the required tables, implementing Read, Create, Write, and Delete ACLs, and finally verifying the complete implementation.